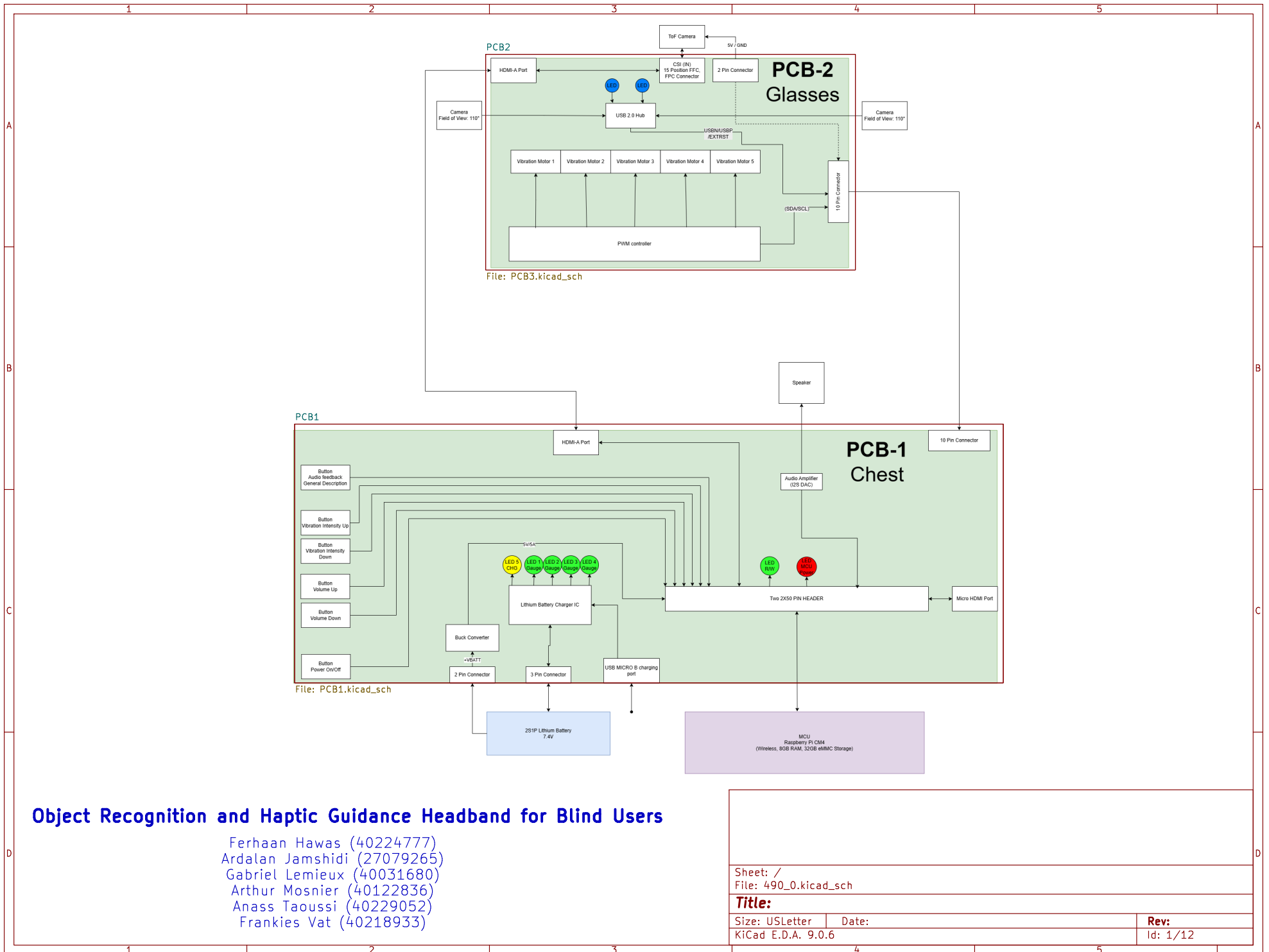




Object Recognition and Haptic Guidance Headband for Blind Users

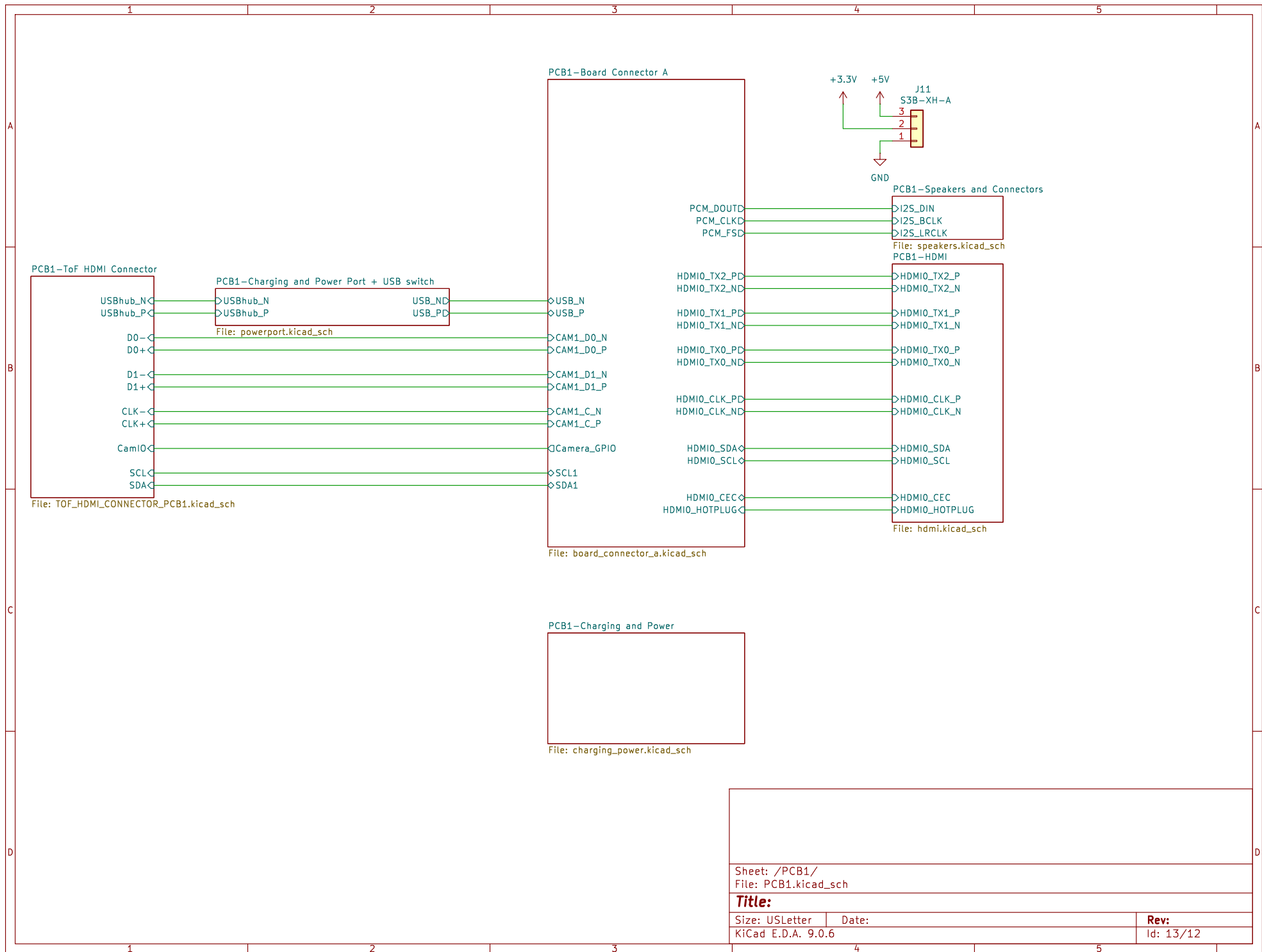
Ferhaan Hawas (40224777)
Ardalan Jamshidi (27079265)
Gabriel Lemieux (40031680)
Arthur Mosnier (40122836)
Anass Taoussi (40229052)
Frankies Vat (40218933)

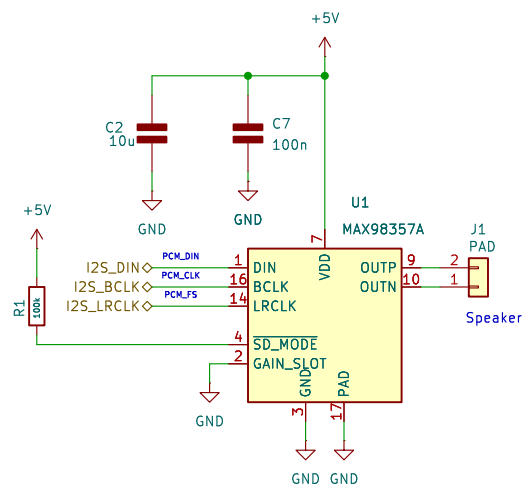
Sheet: /
File: 490_0.kicad_sch

Title:

Size: USLetter Date:
KiCad E.D.A. 9.0.6

Rev:
Id: 1/12





Sheet: /PCB1/PCB1-Speakers and Connectors/
File: speakers.kicad_sch

Title:

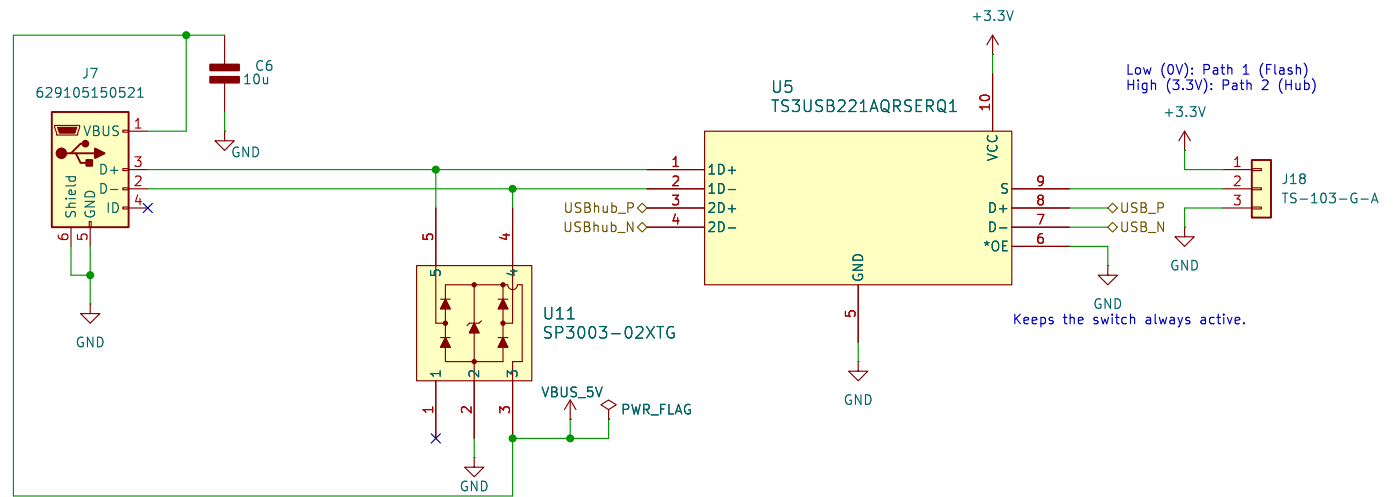
Size: USLetter

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KiCad E.D.A. 9.0.6

Id: 2/12



Sheet: /PCB1/PCB1-Charging and Power Port + USB switch/
File: powerport.kicad_sch

Title:

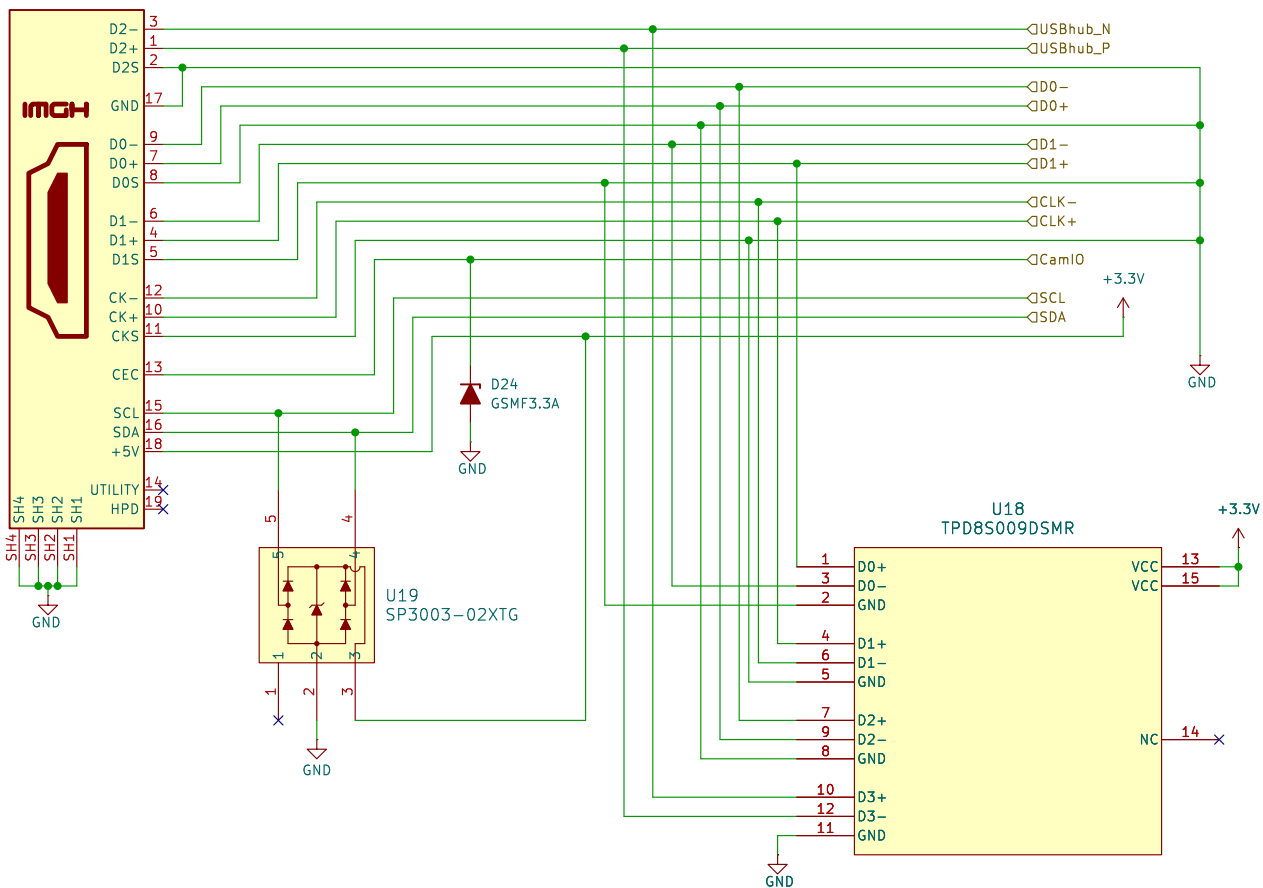
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Id: 11/12

J12
2000-1-1-20-00-BK



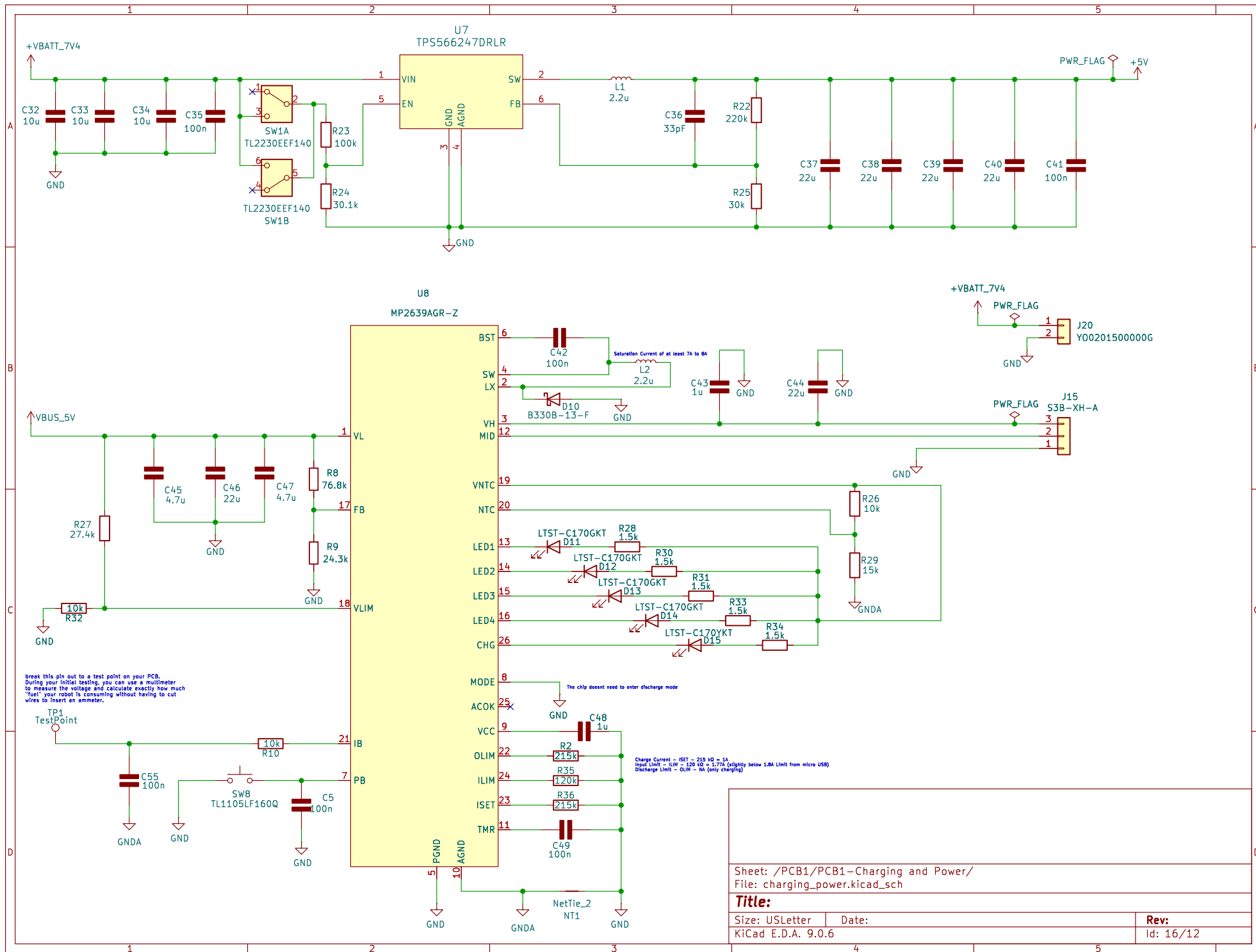
LAYOUT NOTE: > ROUTE MIPI DIFFERENTIAL PAIRS STRICTLY THROUGH SP3003 PADS.
DO NOT USE STUBS. MAINTAIN 100-OHM IMPEDANCE.

Sheet: /PCB1/PCB1-ToF HDMI Connector/
File: TOF_HDMI_CONNECTOR_PCB1.kicad_sch

Title:

Size: USLetter Date:
KiCad E.D.A. 9.0.6

Rev:
Id: 12/12



disable the Ethernet controller in software. Even if the pins are disconnected, the internal hardware is still powered by default.

In /boot/config.txt (or /boot/firmware/config.txt), add:

```
Plaintext
dtparam=eth0=off
```

For Flashing (Device Mode): The CM4 ignores this pin when nRPIBOOT is active, but keeping it floating is the "safe default" for device mode.
For Cameras (Host Mode): force the CM4 to be a Host using software. Add this line to config.txt file after flashing the OS:
dtoverlay=dwc2,dr_mode=host

A

B

C

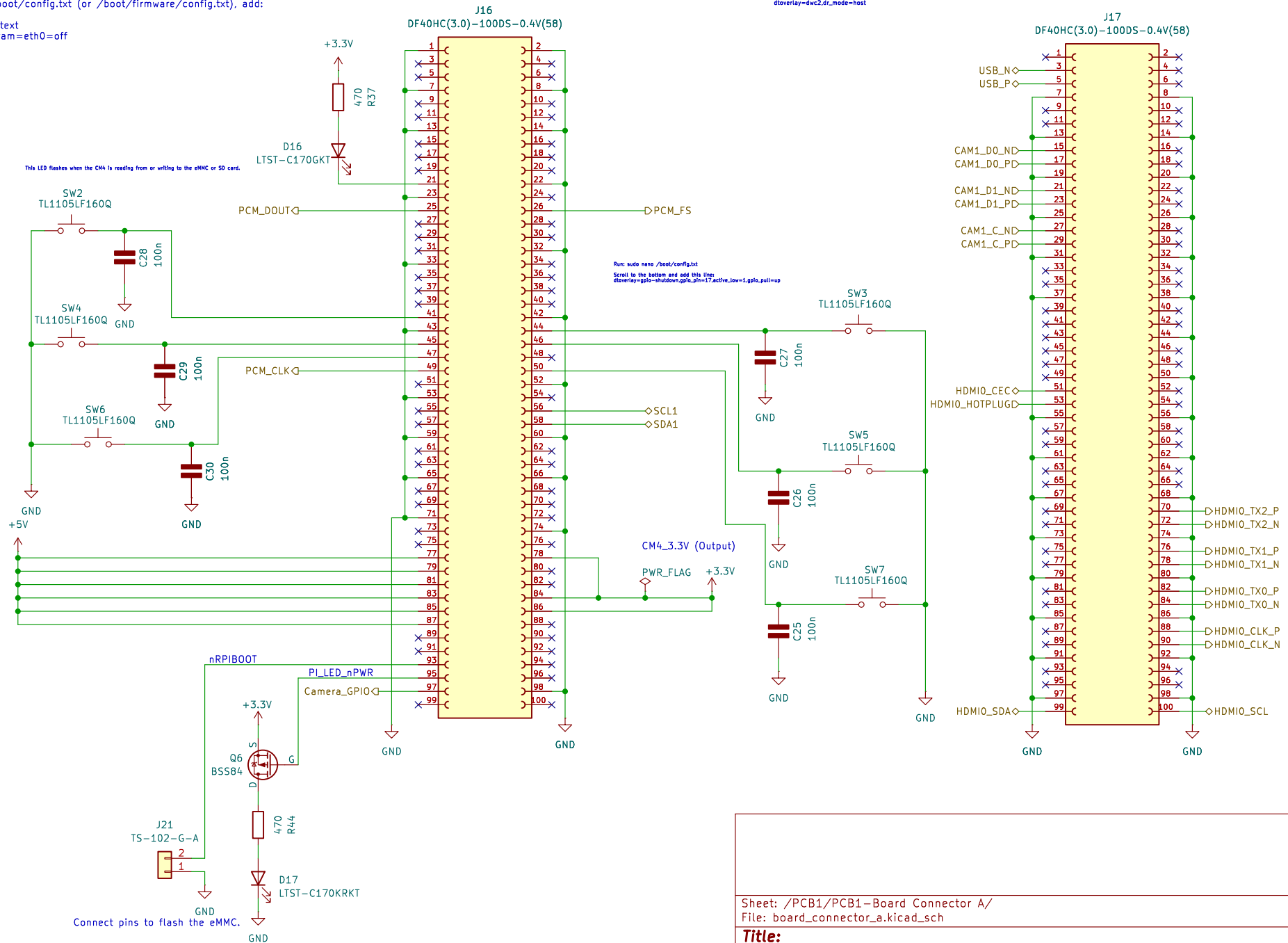
D

A

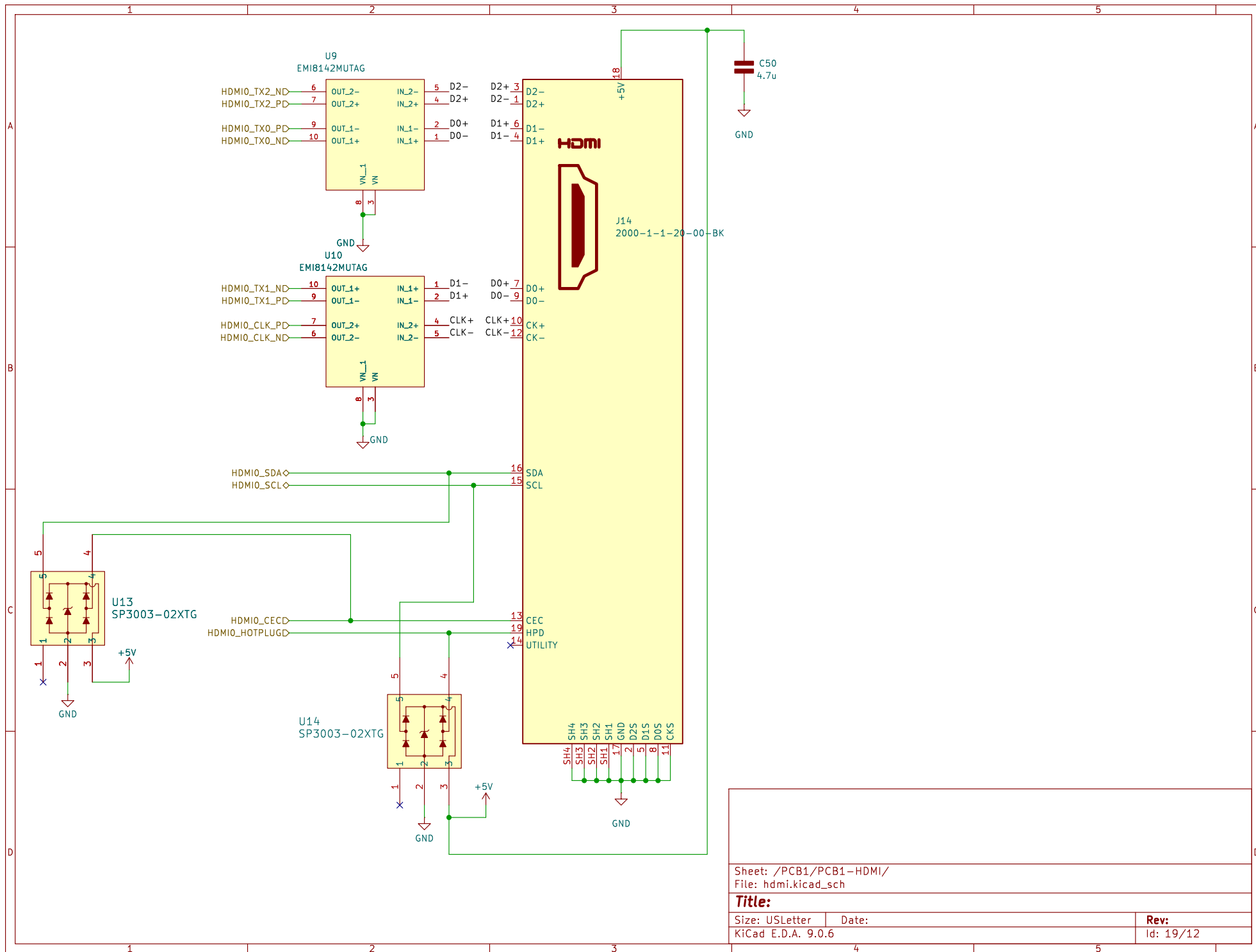
B

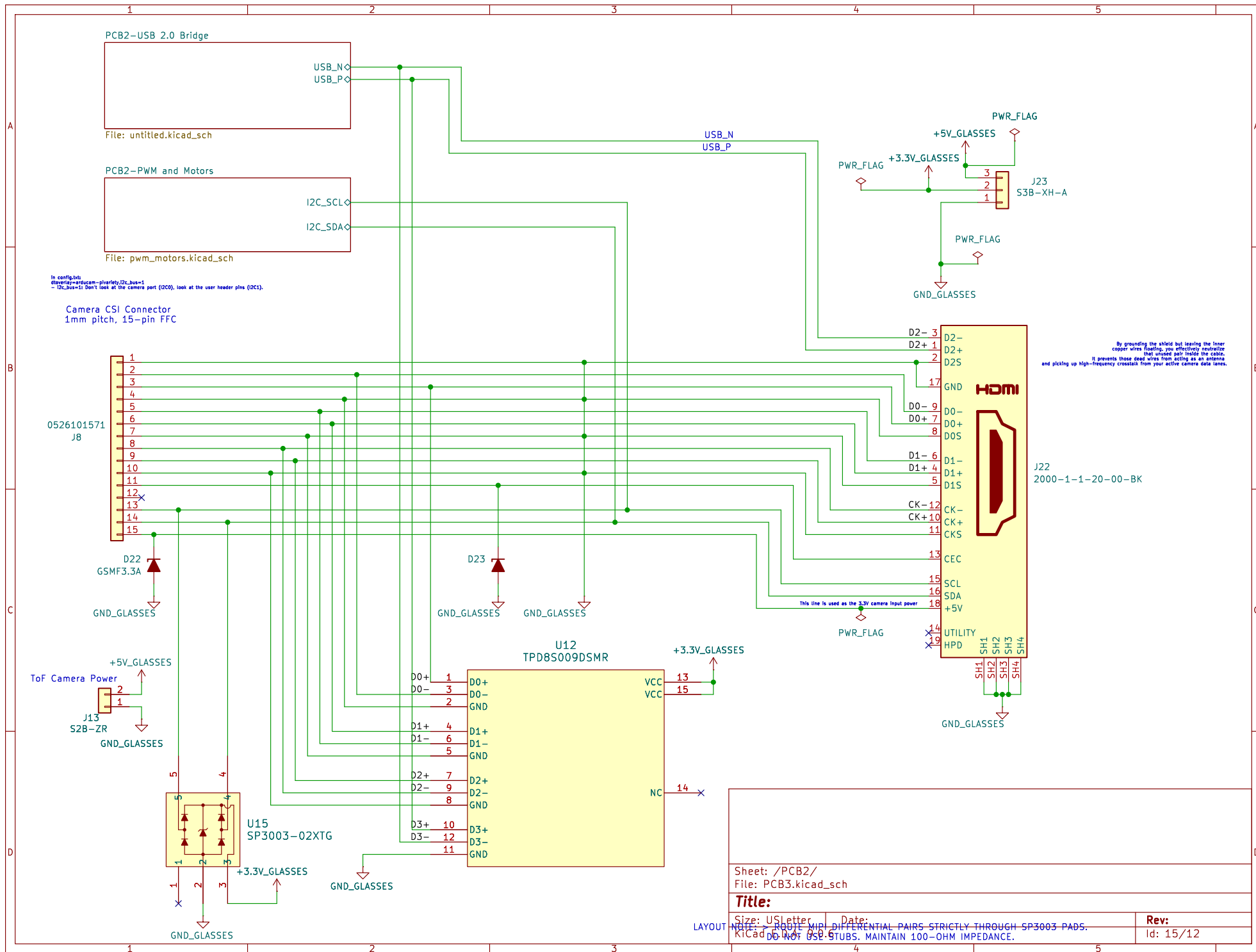
C

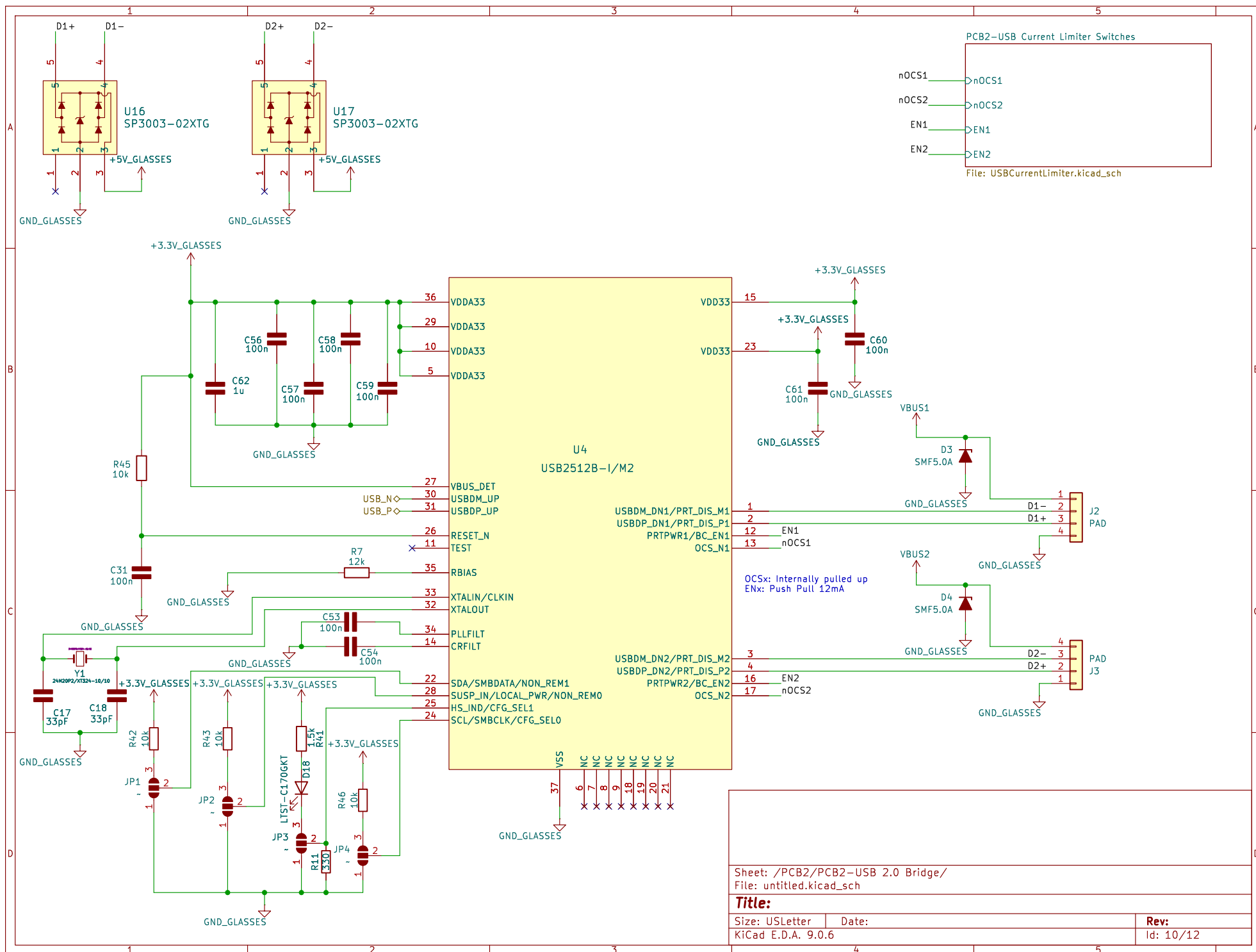
D



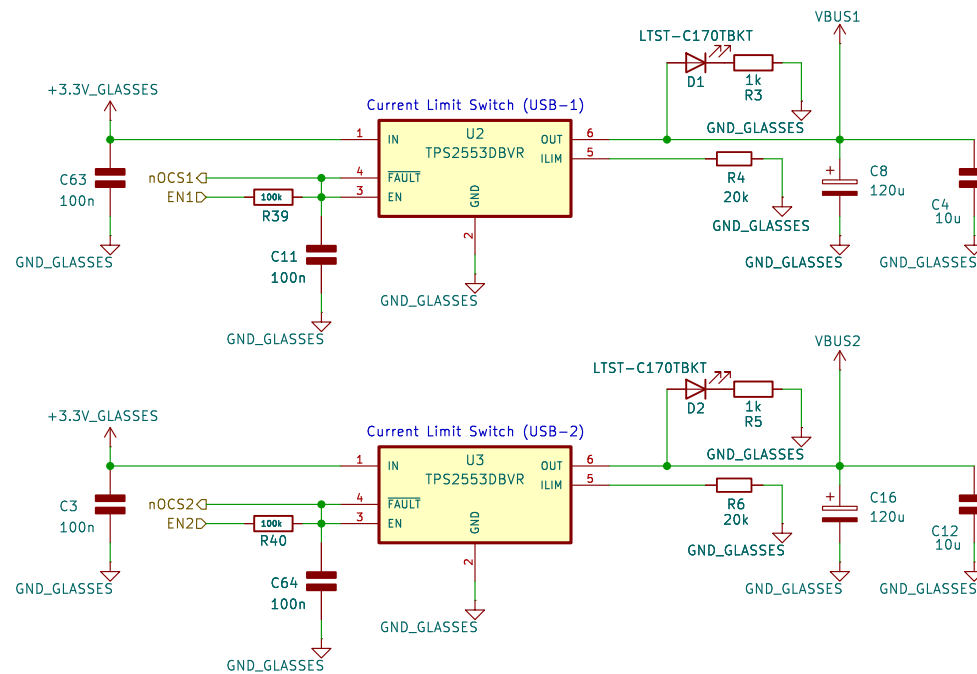
Sheet: /PCB1/PCB1-Board Connector A/ File: board_connector_a.kicad_sch		
Title:		
Size: USLetter	Date:	Rev:
KiCad E.D.A. 9.0.6		Id: 18/12







auto-retry functionality and the ability to enable or disable with an external logic signal.
 Figure 28 shows how an external logic signal can drive EN through RFAULT and maintain auto-retry functionality.
 The resistor-capacitor time constant determines the auto-retry time-out period.



Sheet: /PCB2/PCB2-USB 2.0 Bridge/PCB2-USB Current Limiter Switches/
 File: USBCurrentLimiter.kicad_sch

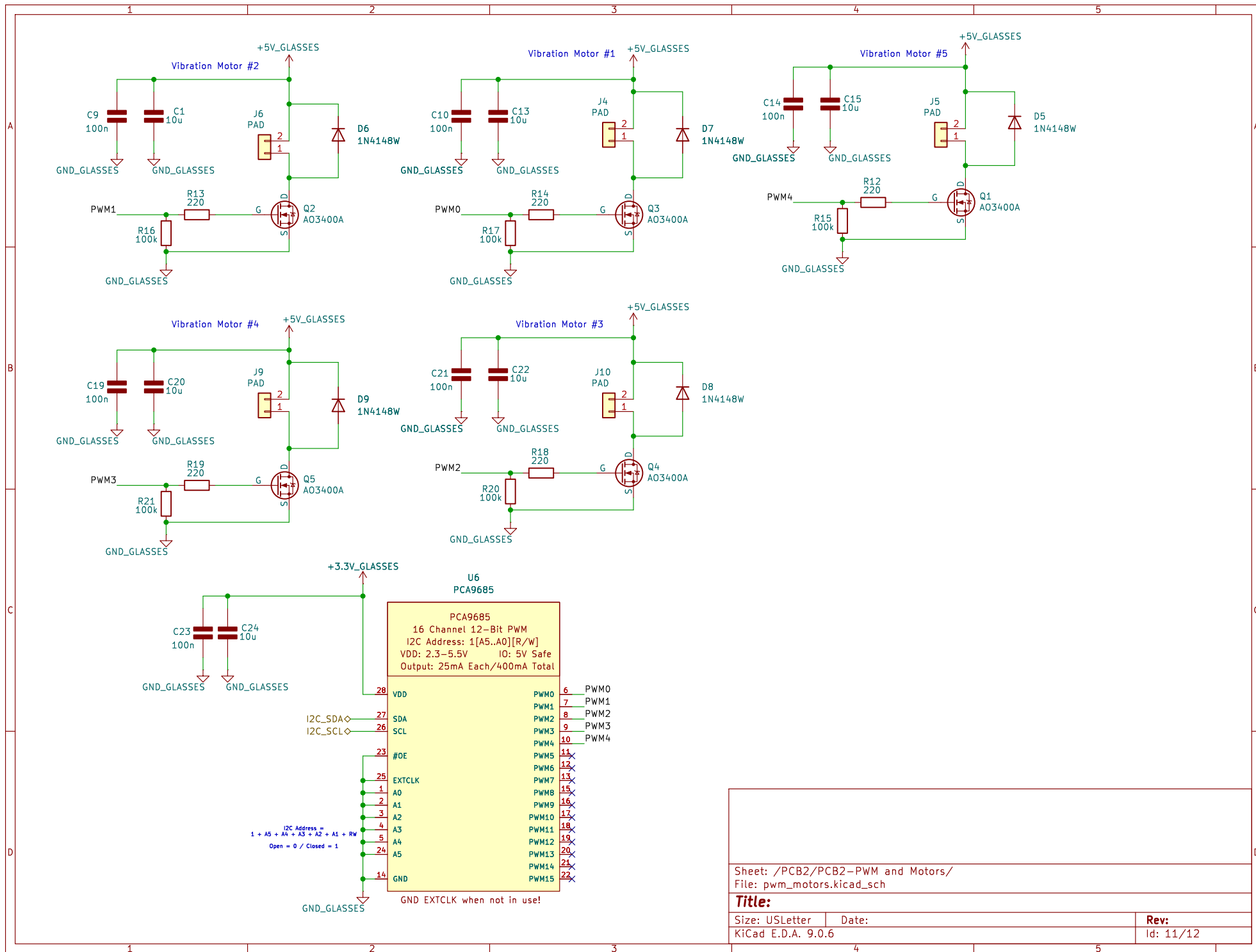
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Size: USLetter
 KiCad E.D.A. 9.0.6

Date:

Rev:

Id: 11/12



Sheet: /PCB2/PCB2-PWM and Motors/
File: pwm_motors.kicad_sch

Title:

Size: USLetter Date:
KiCad E.D.A. 9.0.6

Rev:
Id: 11/12